The Clock That Lies: Why Does GPS Need Einstein's Relativity?

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Imagine you are on an unfamiliar street, and you pull out your phone hoping to see exactly where you are. Within moments, your phone is able to determine your position to a high degree of accuracy — despite the fact that your phone is unable to "see" you from space, relying on signals from satellites around the world which allow your phone to determine your position relative to the satellites.

While commonplace today, this ability actually relies upon one of the most surprising implications of a branch of modern physics: relativity.

GPS not only relies on the satellites, computers, and signals that you would expect. It also relies on another key component: the ability to measure time. And time, as it turns out, is not as absolute as you might previously believed, as shown by Einstein's theories of relativity, which state that the rate at which time passes is not necessarily universal and can be influenced by motion and gravitational effects. These effects are generally miniscule, but GPS' remarkable precision demanded that these effects be taken into account.

How Does GPS Find Your Position?

The Global Positioning System, or GPS, is a collection of satellites that orbit Earth, constantly sending signals to the ground in the form of radio waves. Information encoded in these signals allows GPS receivers on the ground to calculate their position by measuring the amount of time it takes for these signals to reach them.

This works because signals sent via electromagnetic waves have a fixed propagation speed. The time delay between the signal being sent and received, multiplied by this speed, gives a measure of the distance between the receiver and the satellite. The amount of distance can then be converted into position information based on the location of these satellites, using a set of equations like the one below:

$$d = c \cdot \Delta t$$

Where d is the distance, c is the speed of light, and Δt is the time taken for the signal to reach the receiver. Based off of three such distances, the receiver can estimate its position to a high degree of accuracy.

This, however, involves very precise time estimation — as even a minute delay would throw off the estimation of the distance. Because light travels at approximately 300 meters per microsecond (1 μs), an uncorrected timing discrepancy of even one microsecond leads to a positioning error of 300 meters on the ground.

This Brings Us to Einstein

The first issue involves something called time dilation, a result of Einstein's Special Theory of Relativity from 1905.

As a result of time dilation, two observers who move relative to each other will see that the other observer's clock is running slower. The effect is only significant at speeds close to the speed of light, so it is not going to happen if you are simply walking or riding in a bus.

The GPS receiver is much more extreme, since the satellites are whizzing around the earth at incredible speeds, which results in an appreciable amount of time dilation. This causes the clocks in the satellites to actually run slower than the clocks on earth. This can be seen in this equation:

$$\gamma = 1 / \sqrt{(1 - v^2 / c^2)}$$

Where v is the relative speed, and c is the speed of light. Special Relativity alone causes satellite clocks to tick slower than clocks on Earth by about 7 microseconds per day.

General Relativity and Gravitational Dilation

In 1915, Einstein published his General Theory of Relativity. Among many results, this provided an insight into the relationship between matter and time which had a great effect on physicists' understanding of time. This theory predicted something called gravitational time dilation.

Time also appears to flow at a different rate depending on the strength of the gravitational field experienced by an observer. The GPS satellite experiences a weaker gravitational field as compared to a person on the surface of the earth, and thus the satellite's clock runs slightly faster by about 45 microseconds per day.

Relativistic Effect	**Physical Cause**	**Clock Offset Rate**
Special Relativity	Orbital velocity (~14,000 km/h)	-7 microseconds / day
General Relativity	Weaker gravitational field at 20,200 km	+45 microseconds / day
Net Effect	**Combined Relativistic Drift**	**+38 microseconds / day**

The combined result of the two effects is that the GPS satellite's clock actually runs slightly faster than clocks on Earth by about 38 microseconds per day. Without adjusting for these effects, cumulative location drift would exceed 11 kilometers per day.

Conclusion

Einstein's theories of relativity, far from being abstract or theoretical, actually have real-world applications such as GPS. Behind modern maps lie satellites, atomic clocks, equations, and signals being adjusted based on deep principles of modern physics. What started as an abstract theory nearly a century ago found an application in a modern-day technology used by hundreds of millions of people daily.